SOFTWARE ENGINEERING LAB – 3

Component Modelling & Architectural Pattern Selection

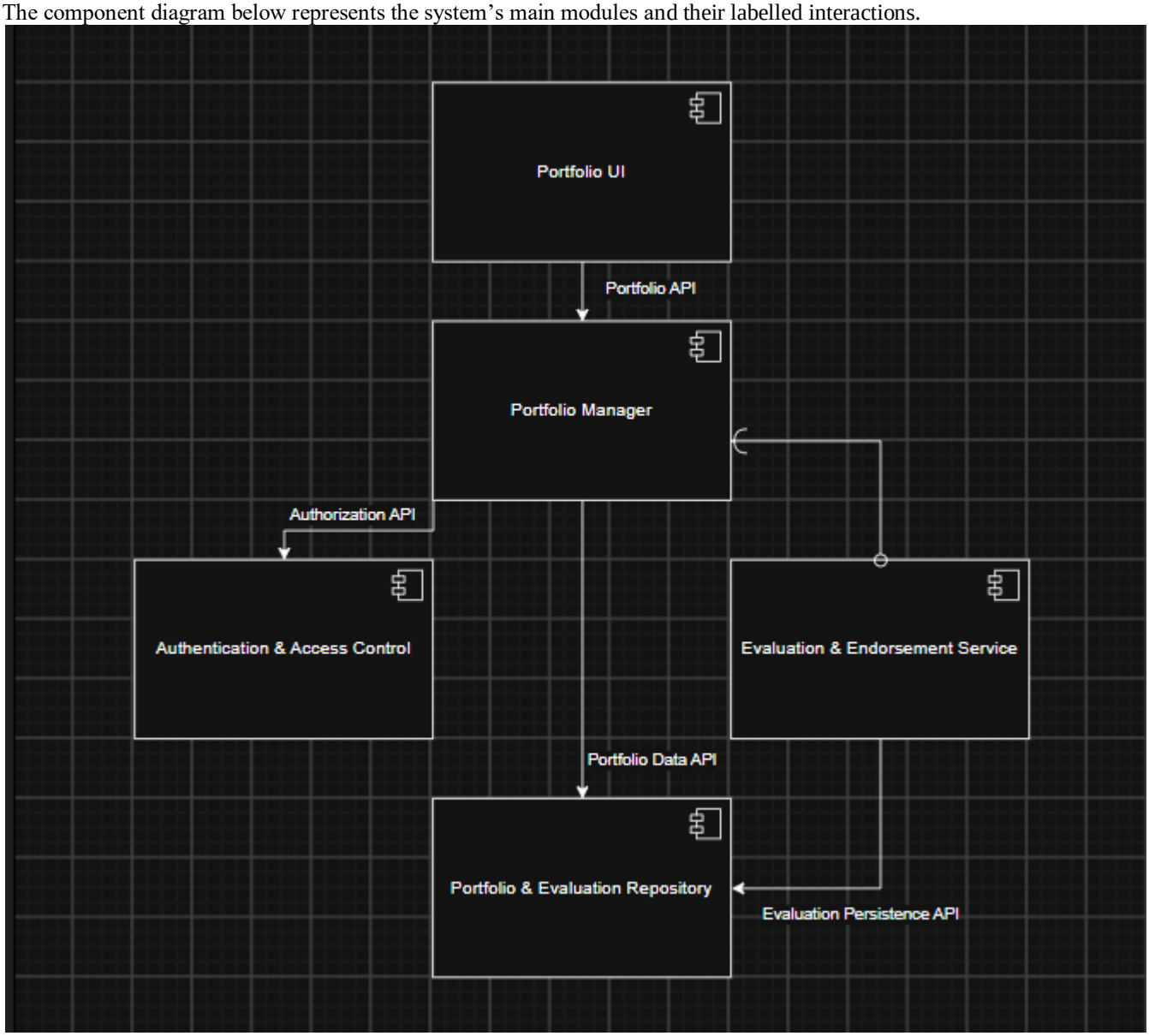
Problem Statement: Student Portfolio Showcase	**Selected Architectural Style:** Layered Architecture
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1. Problem Overview

Student Portfolio Showcase is a platform where students publish project descriptions and links, while peers browse portfolios and leave endorsements. The system must support project/file handling, comment and review workflows, and profile-based access control.

2. UML Component Diagram

The component diagram below represents the system's main modules and their labelled interactions.



3. Component Responsibilities

• **Portfolio UI:** Provides the interface for browsing portfolios, viewing projects, and accessing project-related workflows.

- **Portfolio Manager:** Coordinates portfolio creation and updates, project descriptions and links, browsing requests, and file-handling operations.
- **Authentication & Access Control:** Authenticates users and checks permissions based on user roles and profile access rules.
- **Evaluation & Endorsement Service:** Supports peer endorsements, comments/reviews, and panel-judge project evaluations where applicable.
- **Portfolio & Evaluation Repository:** Persists portfolio/project records, evaluation and endorsement data, and references to uploaded files.

4. Architectural Style Selection and Justification

Layered Architecture is selected because the system can separate presentation, application workflows, security, and data persistence into clearly defined responsibilities. First, this separation makes the system easier to maintain and test: changes to portfolio browsing or review workflows can be handled mainly within the relevant component. Second, it provides a clear structure for the project's distinct workflows—portfolio management, endorsements/reviews, and persistence—without introducing the operational overhead of a distributed microservices design for a project of this scope.

5. Security and Performance Considerations

Security advantage: the Authentication & Access Control component centralizes identity and permission checks, helping ensure that only authorized users can edit portfolios, upload files, or access restricted evaluation data. Uploaded files should also be validated and served only through authorized operations.

Performance benefit: separating UI handling from portfolio, evaluation, and repository operations allows each component to perform focused work and avoids mixing presentation logic with data operations. Clear interfaces can also reduce unnecessary processing and make bottlenecks easier to identify.

6. Conclusion

The proposed component model gives the Student Portfolio Showcase a modular structure with defined responsibilities and labelled interfaces. Layered Architecture is a suitable choice for the current scope because it supports maintainability, controlled access, and clear handling of portfolio and evaluation workflows.